

ANALYSTLAB AFRICA INTERNSHIP PROGRAMME

HealthConnect Experience Lab — Week 4 Kickoff

Machine Learning Problem Definition & Production System Design Report

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Date: August 2026

Project: HealthConnect Clinic

Document Version: 1.0

Status: Kickoff Deliverable

Mentor Access: Granted

Abstract

This document establishes the foundational Machine Learning Problem Definition and Production System Architecture for the HealthConnect Clinic Experience Lab. Facing high non-attendance rates (48.46% baseline no-show rate), HealthConnect Clinic requires an automated, predictive, and actionable system to mitigate appointment missed slots, optimize provider schedule utilization, and improve patient support. We formulate a supervised binary classification task $P(\text{No-Show} \mid X)$, evaluate data quality across 5,000 appointment records, present a candidate feature taxonomy across 6 domain categories, define strict data leakage boundaries, and detail an end-to-end production ML system design incorporating real-time inference microservices, automated multi-channel intervention workflows, MLOps monitoring, and HIPAA-compliant governance.

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1 Executive Summary & Project Context

1.1 Business Scenario

HealthConnect Clinic is an outpatient healthcare provider managing appointment-based clinical consultation and diagnostic services. The clinic suffers from severe operational bottlenecks driven by patient appointment non-attendance. When patients miss scheduled appointments without prior notice, clinical capacity remains idle, patient care continuity is disrupted, and healthcare overhead escalates.

1.2 Central Project Question

How can HealthConnect Clinic leverage data, machine learning, and Generative AI to accurately predict appointment no-shows, deliver proactive patient engagement, and maximize clinical slot utilization?

1.3 Project Transition & Week 4 Scope

During Weeks 1–3, foundational data manipulation and classification modeling techniques were executed on telecom churn data. Beginning in Week 4, the internship transitions to the HealthConnect Experience Lab. Week 4 serves as the critical problem formulation and architectural design phase.

2 Comprehensive Data Quality & Suitability Assessment

2.1 Dataset Overview

The primary analytical resource is `HealthConnect_Appointment_Data.csv`, containing 5,000 synthetic anonymized appointment records across 1,696 unique patients (`patient_id`).

Table 1: HealthConnect Dataset Variable Breakdown & Missingness Profile

Variable Name	Data Type	Missing Count (%)	Role in ML Pipeline	Notes / Mechanism
appointment_id	String	0 (0.0%)	Primary Key	Excluded from modeling
patient_id	String	0 (0.0%)	Grouping Key	Used for GroupKFold sp
gender	Categorical	0 (0.0%)	Feature	Female, Male, Prefer not
age	Integer	0 (0.0%)	Feature	Range: 18–80 years
age_group	Categorical	0 (0.0%)	Feature	6 derived age bands
appointment_type	Categorical	0 (0.0%)	Feature	4 appointment categories
booking_date	Datetime	0 (0.0%)	Temporal	Lead time computation
appointment_date	Datetime	0 (0.0%)	Temporal	Target schedule date
appointment_day	Categorical	0 (0.0%)	Feature	Day of week
appointment_time	Categorical	0 (0.0%)	Feature	Morning, Afternoon, Eve
booking_lead_days	Integer	0 (0.0%)	Feature	Days between booking &
previous_appointments	Integer	0 (0.0%)	Feature	Patient historical count
previous_no_shows	Integer	0 (0.0%)	Feature	Patient historical no-show
reminder_sent	Categorical	0 (0.0%)	Feature	Binary (Yes / No)
reminder_channel	Categorical	1,366 (27.3%)	Feature	NMAR (Missing when Se
distance_to_clinic_km	Float	90 (1.8%)	Feature	MCAR (Median Imputat
waiting_time_minutes	Float	60 (1.2%)	Excluded	MAR (Data Leakage Ris
appointment_outcome	Categorical	0 (0.0%)	Target Source	Attended, No-Show, Can

2.2 Target Outcome Class Distribution

Analysis of `appointment_outcome` across the 5,000 records reveals:

- **No-Show:** 2,423 records (48.46%)
- **Attended:** 2,314 records (46.28%)
- **Cancelled:** 263 records (5.26%)

2.3 Missing Value Mechanisms & Preprocessing Strategy

1. **reminder_channel (27.32% Missing):** Not Missing At Random (NMAR). Missing values correspond exactly to appointments where no reminder was dispatched. Imputation strategy: assign explicit category "None_Sent".
2. **distance_to_clinic_km (1.80% Missing):** Missing Completely At Random (MCAR). Imputation strategy: Group-median imputation by patient postal region / age group.
3. **waiting_time_minutes (1.20% Missing):** Data Leakage Safeguard. Because waiting time is measured post-arrival, utilizing it prior to appointment execution would introduce severe target leakage. Excluded entirely from training features.

2.4 Longitudinal Structure & Cross-Validation Grouping

The 5,000 appointments span 1,696 unique patients (average 2.95 appointments/patient). Standard random K-Fold cross-validation would leak individual patient behavioral traits between training and validation splits. Therefore, model validation must employ **GroupKFold CV** grouped by `patient_id` or a strict **Temporal Train-Test Split** based on `booking_date`.

3 Machine Learning Problem Definition & Formulation

3.1 Supervised Binary Classification Framing

We frame the appointment attendance prediction task as a supervised binary classification problem. For each scheduled appointment i , we predict the probability of non-attendance:

$$P(Y_i = 1 \mid X_i) = \sigma(f(X_i)) \quad (1)$$

where $Y_i = 1$ if the appointment results in a **No-Show**, $Y_i = 0$ if **Attended**, and X_i represents the vector of pre-appointment features available $T - 48$ hours prior to schedule time.

3.2 Operational Handling of Cancellations

Cancelled appointments (5.26%) are filtered out during model training. Operationally:

- **Cancellations** reflect proactive patient communication, enabling clinic reception to re-book the open slot immediately.
- **No-Shows** reflect unannounced absences resulting in unrecoverable idle clinician capacity.

Hence, the primary ML system targets binary discrimination between Attended ($Y = 0$) and No-Show ($Y = 1$).

3.3 Proposed Candidate Feature Taxonomy

1. **Patient Demographics:** age, gender, age_group.
2. **Appointment Attributes:** appointment_type, appointment_day, appointment_time.
3. **Temporal & Lead Time Metrics:** booking_lead_days, lead time binned windows (0-3d, 4-7d, 8-14d, 15-30d, 31-60d, 60d+).
4. **Historical Behavioral Signals:** previous_appointments, previous_no_shows, engineered feature $\text{historical_noshow_ratio} = \frac{\text{previous_no_shows}}{\max(\text{previous_appointments}, 1)}$.
5. **Communication Channels:** reminder_sent, reminder_channel (SMS, WhatsApp, Email, None_Sent).
6. **Geographical Logistics:** distance_to_clinic_km.

4 End-to-End Production Machine Learning System Design

4.1 High-Level System Architecture

The production ML prediction platform consists of five modular microservices:

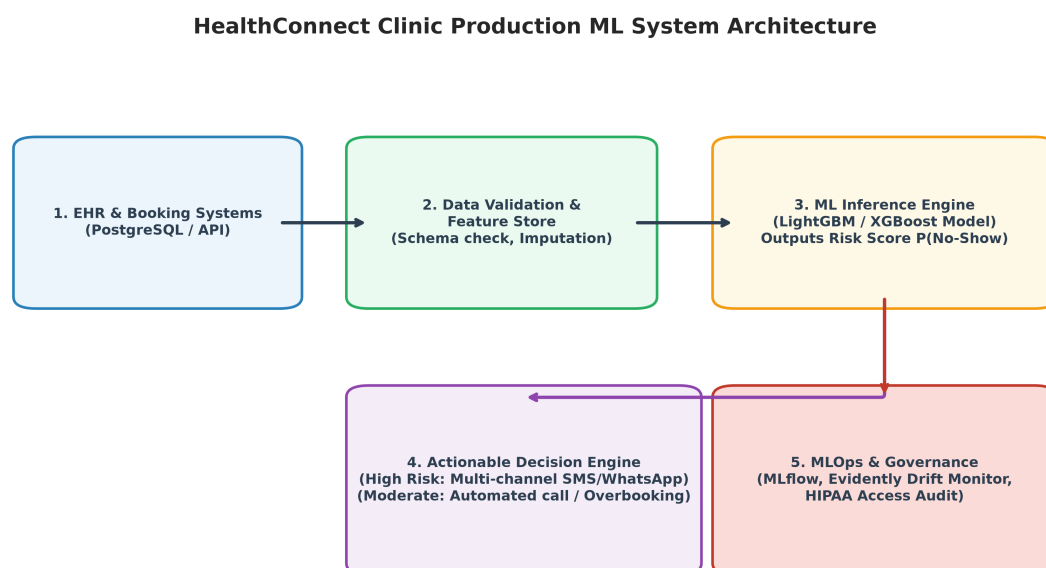


Figure 1: Production Machine Learning System Architecture for HealthConnect Clinic

1. **EHR Ingestion Pipeline:** Synchronizes upcoming appointments daily from PostgreSQL EHR database $T - 48$ hours prior to scheduled date.
2. **Feature Store & Schema Validator:** Validates data schema using Great Expectations, imputes missing values, and fetches historical patient aggregations.
3. **ML Inference Service:** Containerized FastAPI service serving calibrated LightGBM / XGBoost models, returning $P(\text{No-Show})$.

4. **Actionable Decision Engine:** Classifies risk tier and triggers automated patient engagement:
 - **High Risk** ($P > 0.70$): Multi-channel SMS + WhatsApp interactive confirmation; flag slot for standby buffer overbooking.
 - **Moderate Risk** ($0.40 \leq P \leq 0.70$): Automated SMS reminder requesting 1-click confirmation or rescheduling.
 - **Low Risk** ($P < 0.40$): Standard single email notification.
5. **MLOps & Drift Monitoring:** Logs inputs, predictions, and actual outcomes to MLflow registry and Prometheus/Grafana dashboards.

4.2 API Schema Specifications

Request Payload (POST /v1/predict-noshow):

```
{
  "appointment_id": "HC-00412",
  "patient_id": "P-0912",
  "age": 42,
  "gender": "Female",
  "appointment_type": "Specialist Consultation",
  "booking_lead_days": 18,
  "previous_appointments": 4,
  "previous_no_shows": 2,
  "reminder_sent": "Yes",
  "reminder_channel": "SMS",
  "distance_to_clinic_km": 14.2
}
```

Response Payload:

```
{
  "appointment_id": "HC-00412",
  "no_show_probability": 0.764,
  "risk_tier": "HIGH_RISK",
  "recommended_action": "TRIGGER_WHATSAPP_CONFIRMATION_AND_BUFFERSLOT",
  "model_version": "v1.2.0-lightgbm"
}
```

5 Reproducibility, MLOps, Governance & Ethics

5.1 Reproducibility Framework

All model artifacts, hyperparameters, pipelines, and evaluation metrics will be version-controlled via DVC (Data Version Control) and tracked using MLflow. Code repositories maintain deterministic random seeds across data splitting, imputation, and model training routines.

5.2 Model Governance & Monitoring

- **Data Drift Monitoring:** Statistical Kolmogorov-Smirnov tests executed weekly via Evidently AI to detect feature drift in `booking_lead_days` or demographic shifts.
- **Concept Drift Monitoring:** Performance metrics (Recall, ROC-AUC, Log-Loss) evaluated weekly against actual attendance logs.

5.3 Ethical Considerations & Fairness

Healthcare predictive models must avoid systemic bias against vulnerable populations. Models will be evaluated for equalized odds and equal opportunity across demographic sub-groups (`age_group`, `gender`, geographical distance). Interventions are non-punitive, providing extra engagement support rather than denying service.

6 Assumptions, Limitations, Risks & Next Steps

6.1 Risk Matrix

Table 2: Project Risk Matrix & Mitigation Strategy

Risk Event	Description	Severity	Probability	Mitigation Strategy
Data Leakage	Post-arrival feature included in pre-appointment model	High	Medium	Exclude <code>waiting_time_minutes</code> ; audit feature store pipelines.
Target Drift	Post-COVID patient behavior alters attendance patterns	High	Medium	Automated weekly drift detection & bi-monthly model retraining.
Class Imbalance	Skewed outcome distributions across specialist clinics	Medium	Low	Cost-sensitive threshold tuning (F_β optimization with $\beta = 2$).
Cold Start	New patients lacking historical appointment records	Medium	High	Population prior imputation + lead time baseline weighting.

6.2 Proposed Focus for Week 5

1. Construct modular scikit-learn feature preprocessing pipelines (One-Hot Encoding, Robust Scaling, Imputation).
2. Engineer behavioral interaction terms (`noshow_ratio`, `lead_time_weeks`).
3. Train and benchmark 5 baseline classifiers: Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, LightGBM.
4. Evaluate performance on recall-focused metrics (F_2 -score, ROC-AUC, PR-AUC).